INCH-POUND

MIL-PRF-81757/15(AS)
1 March 1999

PERFORMANCE SPECIFICATION SHEET

BATTERY, STORAGE, AIRCRAFT, NICKEL-CADMIUM, HIGH RATE,
24-VOLT, 25-AMPERE-HOUR, LOW-MAINTENANCE, VENTED

This specification is approved for use by the Naval Air Systems Command, Department of the Navy, and is available for use by all Departments and Agencies of the Department of Defense.

The requirements for acquiring the product described herein shall consist of this specification sheet and MIL-PRF-81757.

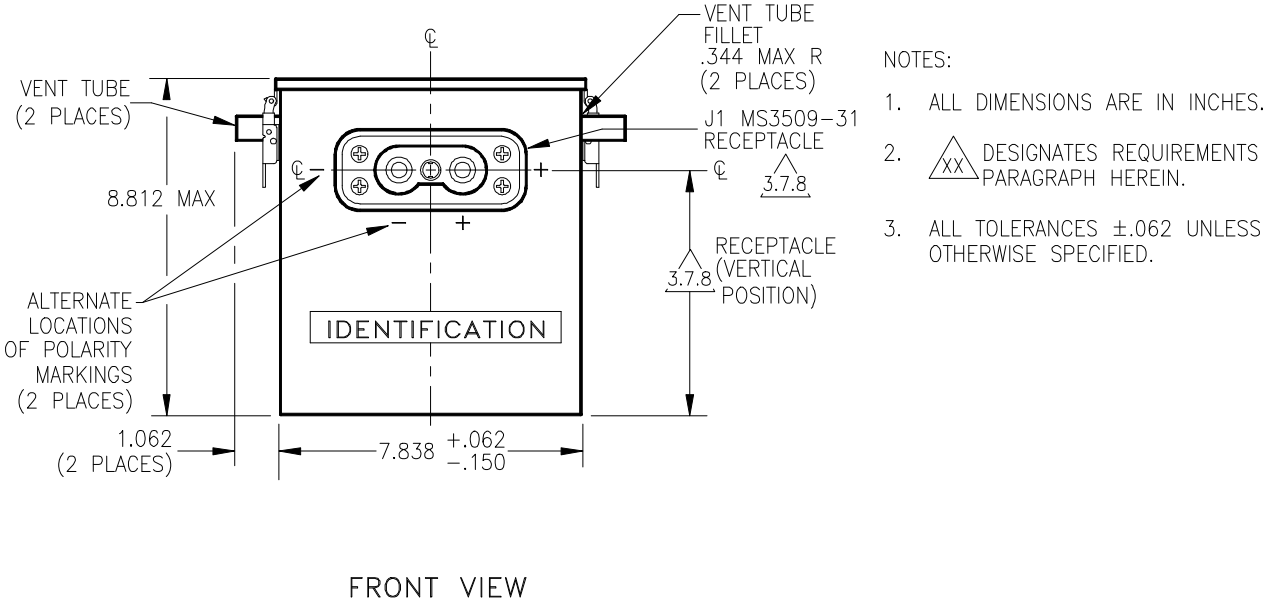


FIGURE 1. Dimensions and configuration for PIN M81757/15-1.

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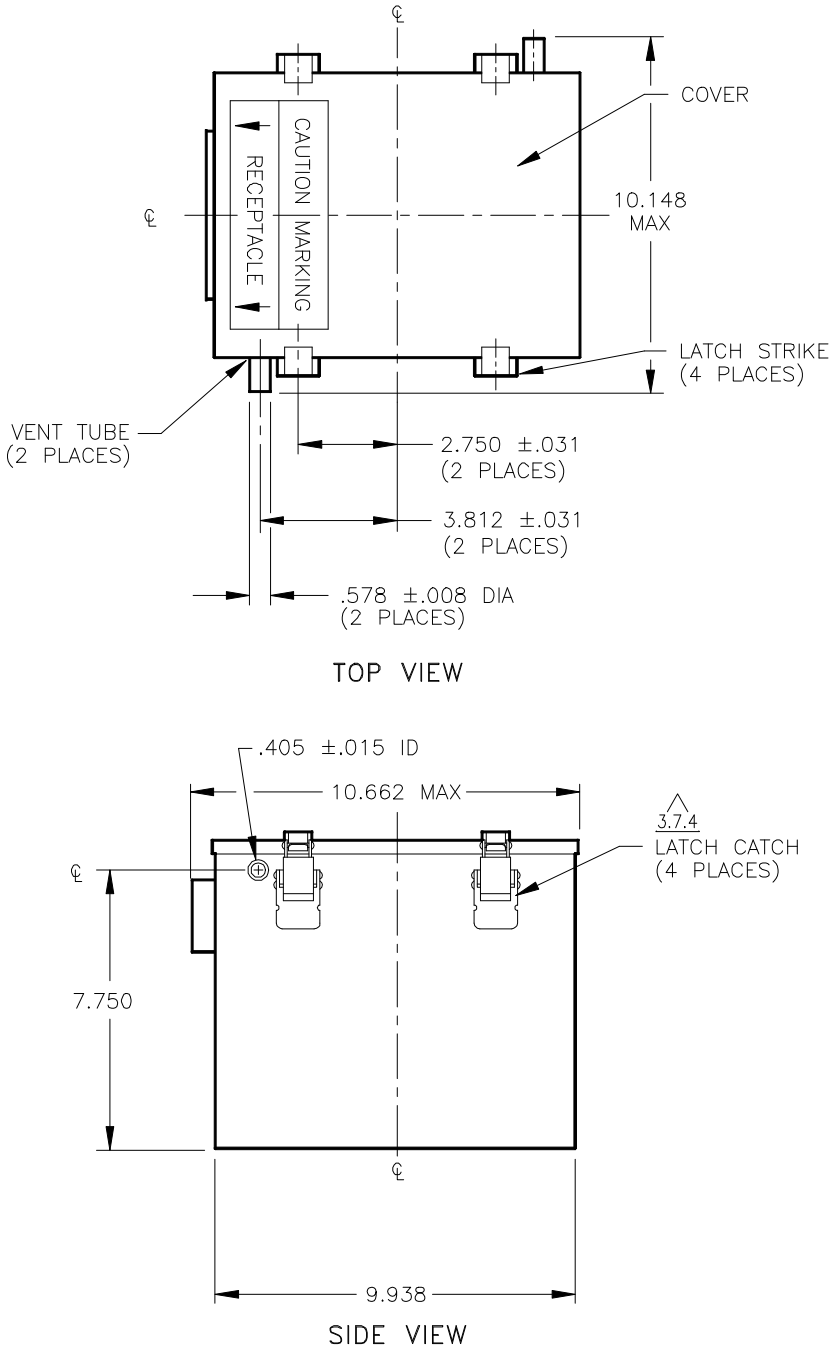
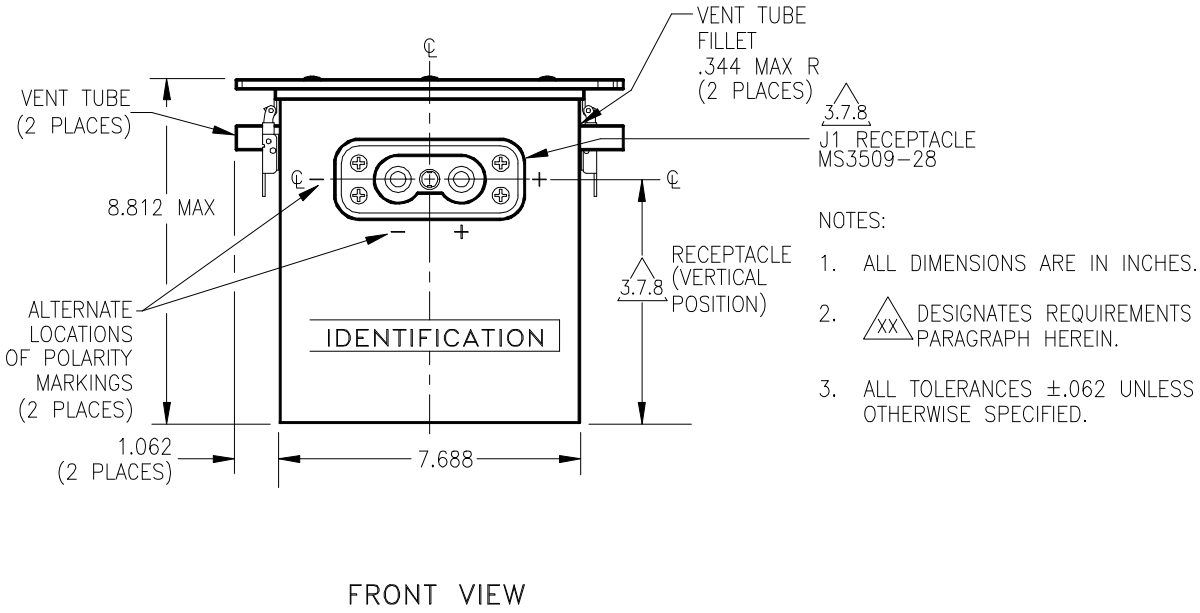


FIGURE 1. Dimensions and configuration for PIN M81757/15-1 - Continued.

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FRONT VIEW

FIGURE 2. Dimensions and configuration for PIN M81757/15-2.

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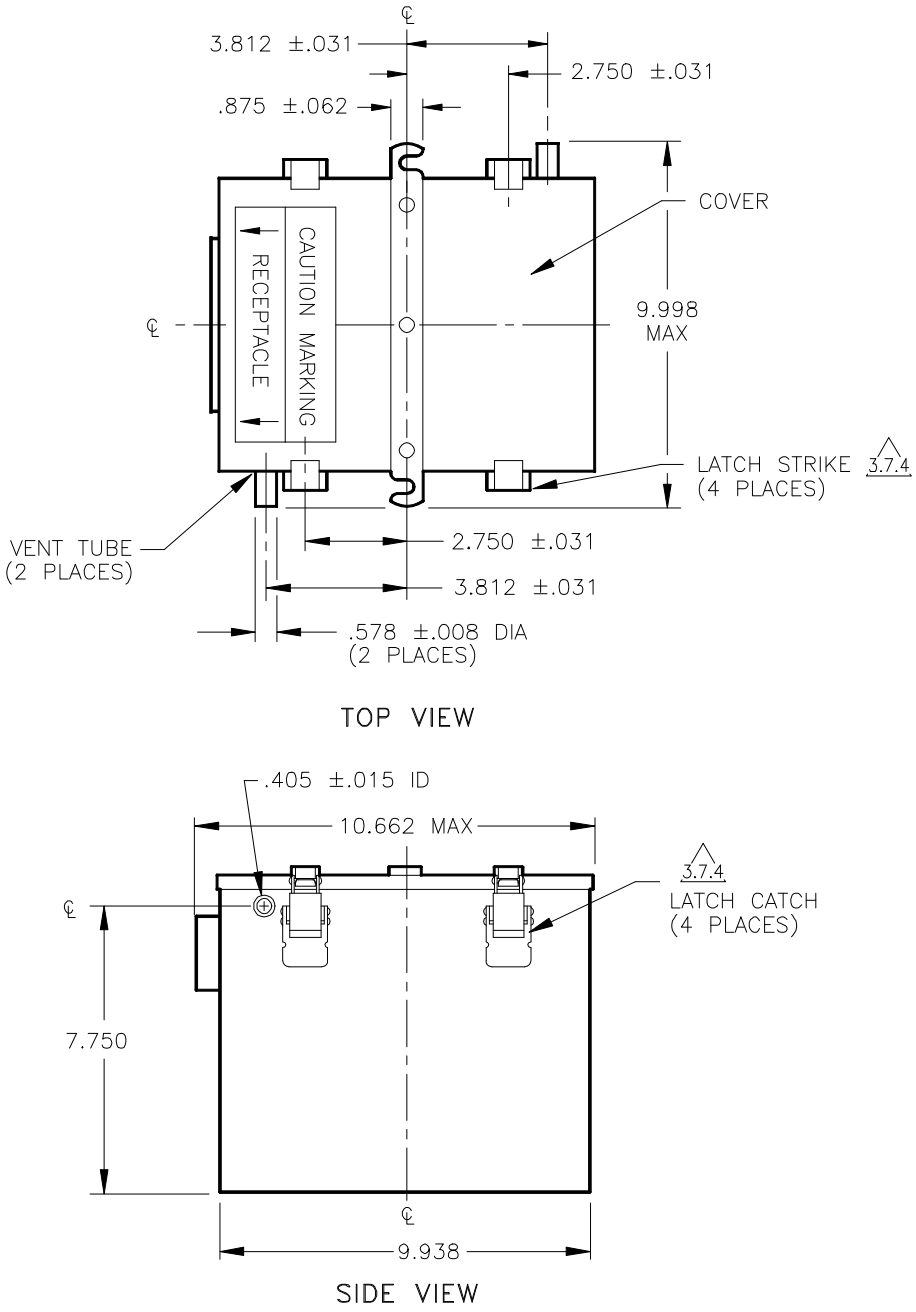
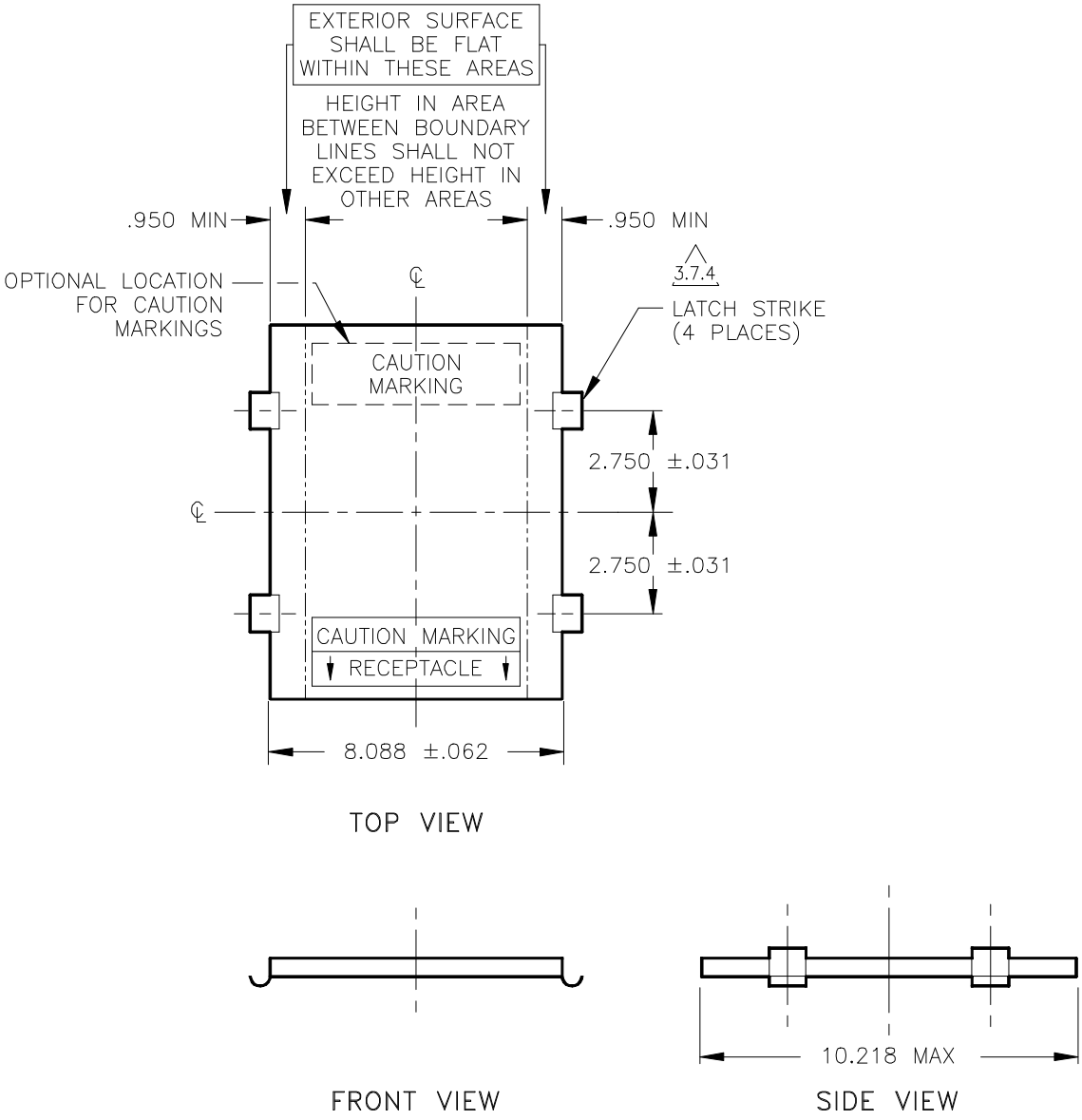


FIGURE 2. Dimensions and configuration for PIN M81757/15-2 - Continued.

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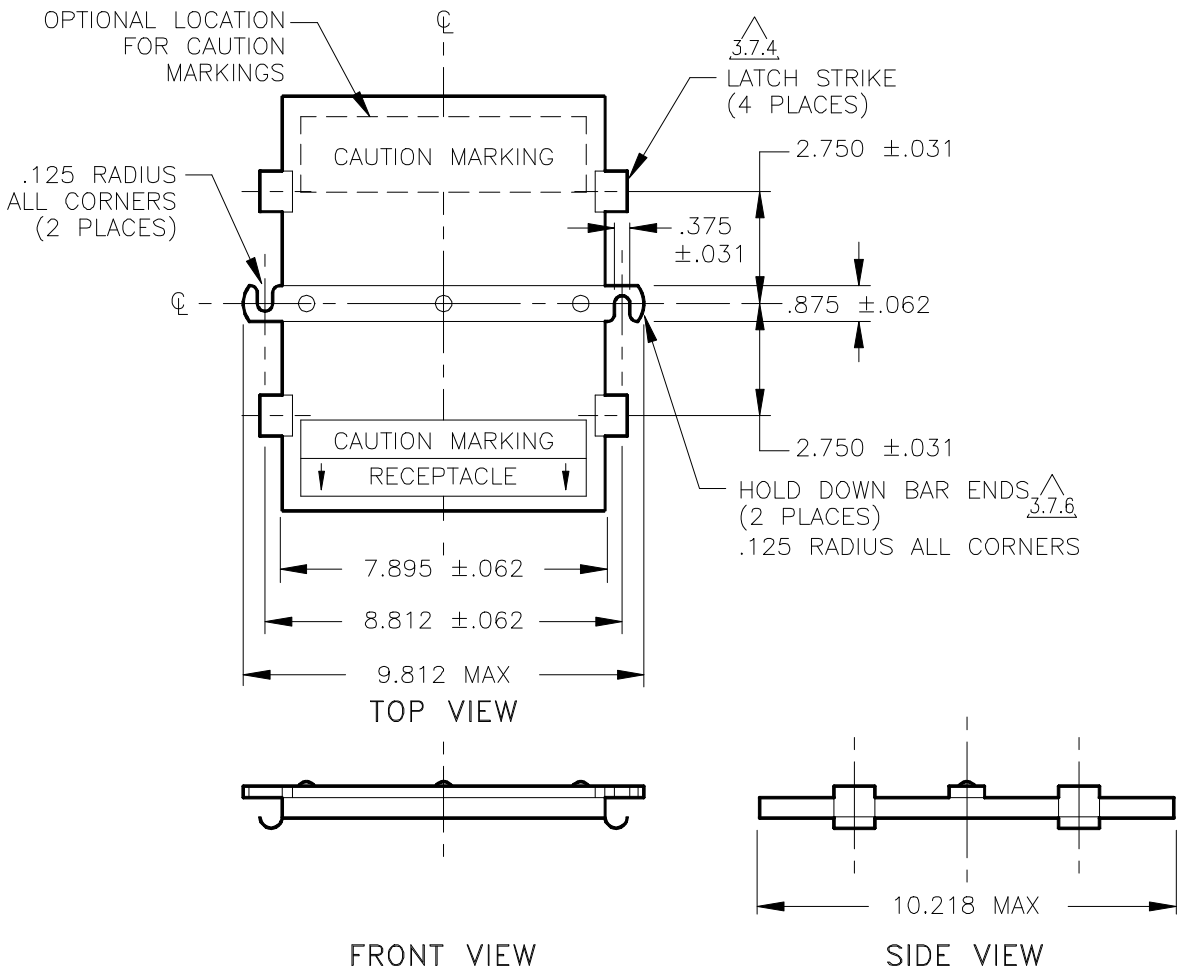


NOTES:

1. ALL DIMENSIONS ARE IN INCHES.
2. XX DESIGNATES REQUIREMENTS PARAGRAPH HEREIN.

FIGURE 3. Cover for PIN M81757/15-1.

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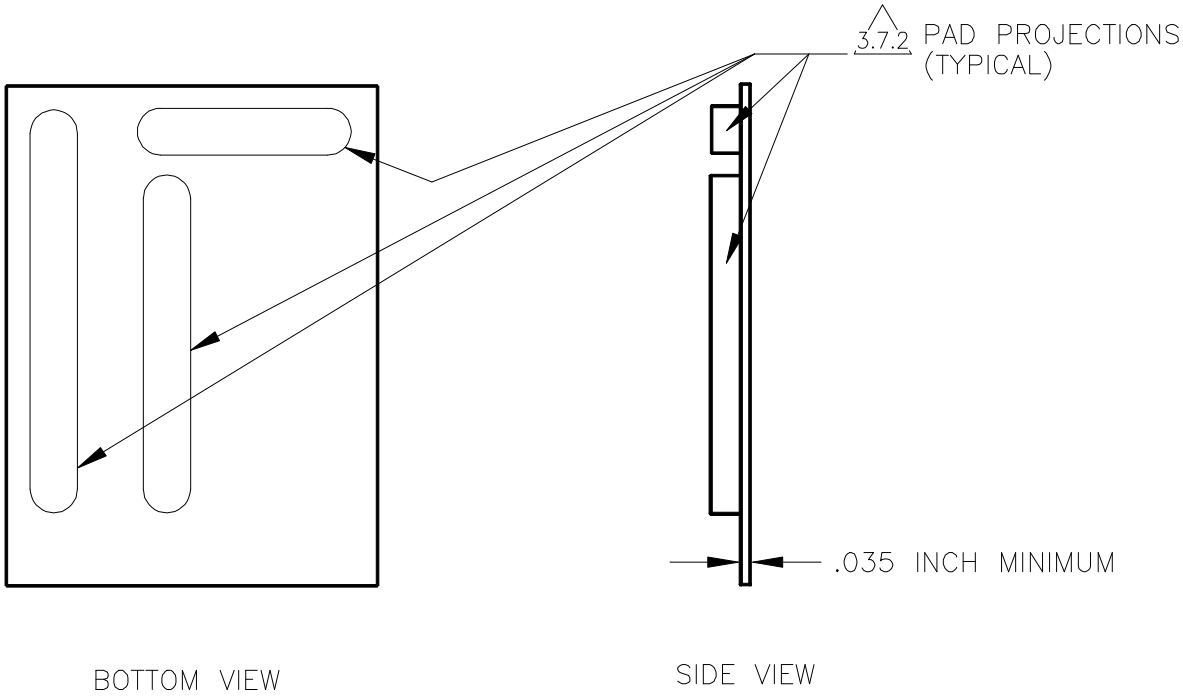


NOTES:

1. ALL DIMENSIONS ARE IN INCHES.
2. XX DESIGNATES REQUIREMENTS PARAGRAPH HEREIN.

FIGURE 4. Cover for PIN M81757/15-2.

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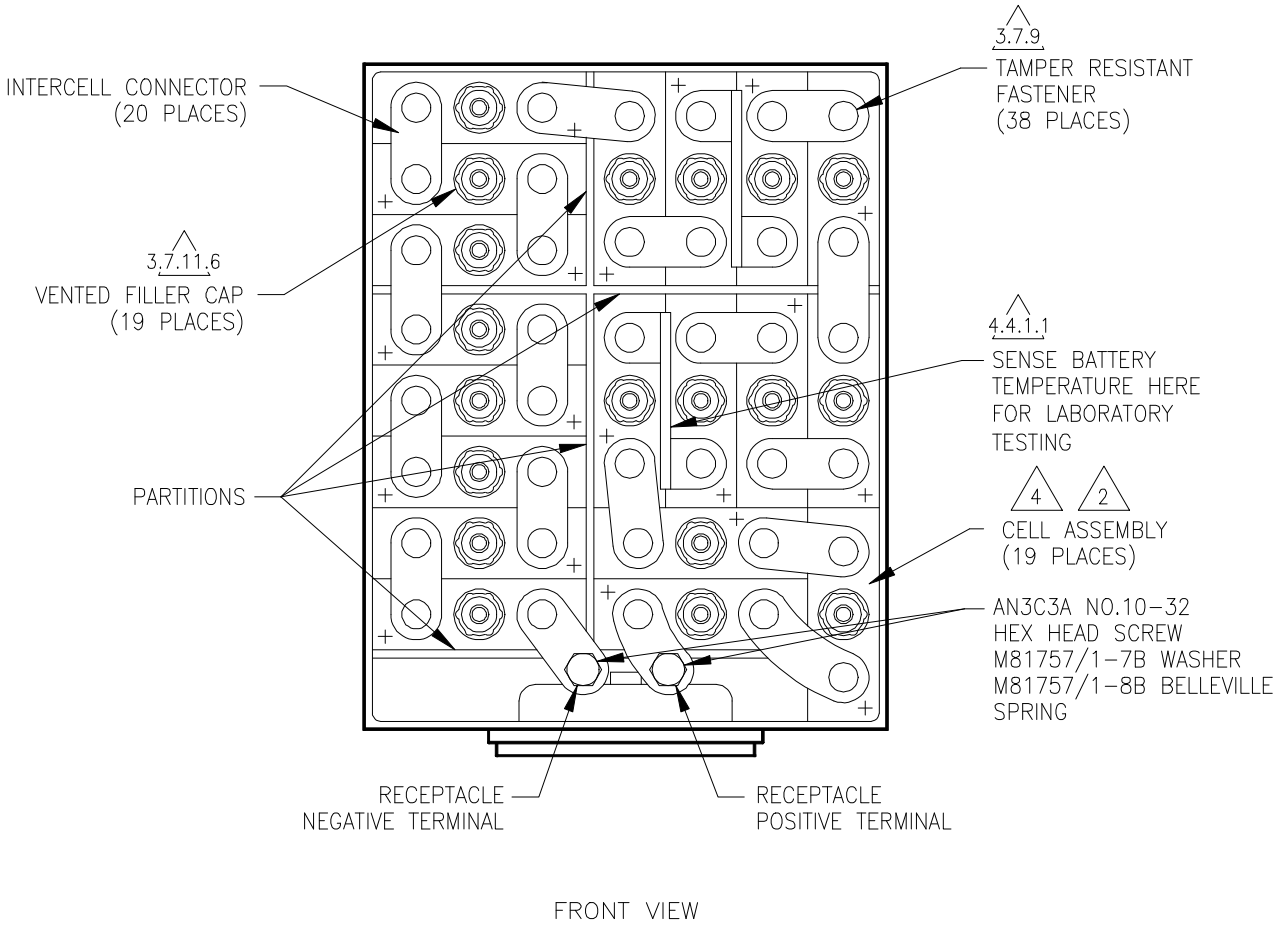


NOTES:

1. THE PAD SHALL COMPLY WITH 3.4, 3.7.2, AND 3.7.3.
2. XX DESIGNATES REQUIREMENTS PARAGRAPH HEREIN.
3. WHEN COVER IS SECURED, THE PROJECTIONS SHALL CONTACT EACH INTERCELL CONNECTOR (FOR PIN M81757/15-1 ONLY) OR EACH VENTED FILLER CAP (FOR PIN M81757/15-2 ONLY).

FIGURE 5. Cover gasket/cell hold-down pad for M81757/15-X.

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FRONT VIEW

NOTE:

1. XX DESIGNATES REQUIREMENTS PARAGRAPH HEREIN.

FIGURE 6. Layout of interior components for PIN M81757/15-X.

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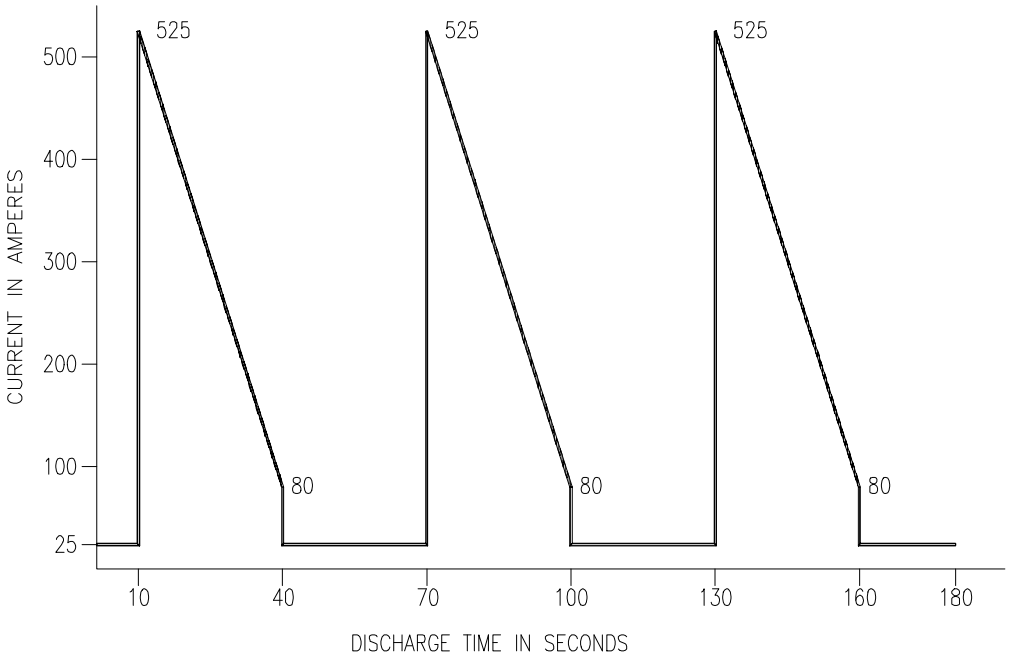


FIGURE 7. Slope discharge profile.

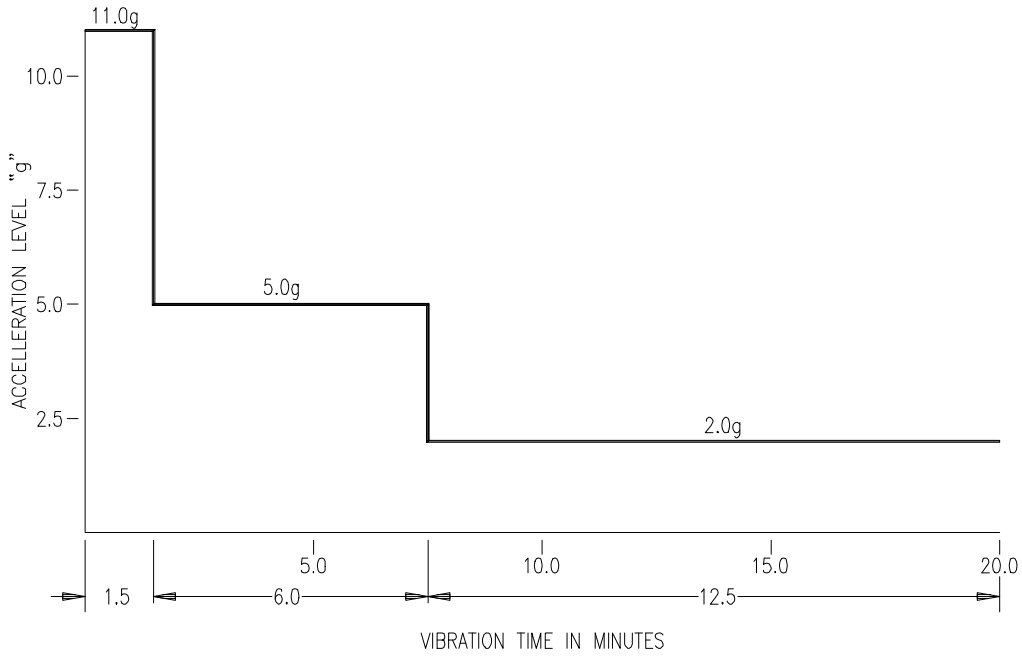


FIGURE 8. Low frequency (17 Hz) vibration.

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NOMENCLATURE:	BATTERY, STORAGE, AIRCRAFT, NICKEL-CADMIUM, HIGH RATE, LOW-MAINTENANCE, VENTED, LIMITED-REPAIR
MILITARY PIN:	M81757/15-X
NATIONAL STOCK NUMBER:	6140-**-***-****
NOMINAL VOLTAGE:	24 VOLTS
MAXIMUM WEIGHT:	***
INITIAL CAPACITY:	29 AH (25 AMPS/70 minutes/77 °F/19 V)
MINIMUM CAPACITY:	25 AH (25 AMPS/1 HR/77°F/20 V)
CONTRACT AND DELIVERY ORDER NUMBER:	***
MANUFACTURED BY:	***
MANUFACTURER'S CAGE:	***
MANUFACTURER'S SERIAL NUMBER:	***
DATE/LOT CODE:	***
DATE FIRST PLACED IN SERVICE:	

FIGURE 9. Marking for M81757/15-X batteries.

	MINIMUM LETTERING HEIGHT
CAUTION	0.500 inch minimum lettering height.
This cover manufactured by CAGE ***** and shall be installed only on a M81757/15-X battery manufactured by same CAGE.	0.250 inch minimum lettering height.
The cover of other M81757/xx batteries shall not be installed on a M81757/15-X battery.	0.250 inch minimum lettering height.
The M81757/15-X battery shall be serviced only in an authorized nickel-cadmium battery shop. The cells shall not be removed.	0.250 inch minimum lettering height.
⇩ RECEPTACLE ⇩	0.500 inch minimum lettering height.

FIGURE 10. Marking for battery caution and receptacle.

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Applications	M81757/15-1	M81757/15-2
Applicable figures herein	Figures 1, 3, and 5 through 11	Figures 2, 4, 5, 6 and 9 through 11
Weight (lbs)	53.5 ± 1.5	52 maximum
Vented filler cap for cells	M81757/2-2	M81757/2-8
Receptacle	MS3509-31	MS3509-28
Battery mounting	No hold-down bar	Has hold-down bar
Special test/requirements	Low frequency vibration slope discharge	None
User aircraft	H-2, H-3	T-2

FIGURE 11. Applicable characteristics for each PIN.

REQUIREMENTS:

1. Part or identifying number. The part or identifying numbers (PINs) of the batteries covered by this specification sheet shall be M81757/15-1 and M81757/15-2, collectively known as M81757/15-X. The M81757/15-X is a limited-repair, low-maintenance, vented battery.

2. Dimensions and weight. The dimensions and weight of the battery shall be as specified in 3.10.8 as modified herein and as shown on figures 1 through 5 and figure 11 contained herein. The exterior dimensions of each cell container shall be identical to the exterior dimensions of PIN M81757/1-4.

3. Capacity. The initial capacity of the battery shall be not less than 29 ampere-hours (Ah). The minimum capacity of the battery shall be not less than 25 Ah. See 3.10.14 as modified herein.

4. Interface requirements between battery and test fixtures. The battery shall contain 19 cells. The cells and other interior components of the battery shall be arranged and positioned as shown on figure 6 herein. The cell terminals shall be vertically and horizontally positioned so that each cell terminal shall mate with its matching electrical connection of the cell scanner

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fixture, PIN 371AS264 and the cell equalization fixture, PIN 371AS265. The fasteners specified on figure 6 are required due to interfaces with the MS3509 receptacle, the cell scanner fixture, and the cell equalization fixture.

5. Battery partition. The battery partition that contains the center row of cells and interfaces with the exterior side wall of cell number 16 (counting from the battery's negative terminal) shall be constructed of thermally conductive material. The height of the partition shall be not less than the height of the plates within cell number 16. The partition's surface adjacent to the side wall of cell number 16 shall be smooth and shall contact the cell's side wall. Spacers or air passages shall not be present at this interface. The partition shall have rigidity and strength to contain cell number 16 for the life of the battery without warping or breaking. The partition shall not distort due to undergoing any or all of the tests herein.

6. MIL-PRF-81757 variance. The battery shall comply with MIL-PRF-81757 except as follows.

6.1 Modify the following paragraphs:

3.7.2 Battery container and cover. Delete the entire text and substitute: "The battery container and cover shall be made of a material that is impervious to salt spray and to an aqueous solution of potassium hydroxide of 1.30 ± 0.04 specific gravity (see 6.10.1) and an alternate electrolyte if used. The container and cover may be constructed of materials that are not identical. The cover may be protected with a material (such as paint, epoxy, or powder coating) that is impervious to salt spray and to an aqueous solution of potassium hydroxide of 1.30 ± 0.04 specific gravity (see 3.8.1 and 6.10.1) and an alternate electrolyte if used. Such protective materials shall not be applied to the battery container except as indicated in 3.8.1. The battery container and cover shall be free of pits, blow holes, rough spots, and other deformations. The dimensions and locations of receptacles, hold-downs, latches (catches and strikes), vent tubes, and battery identification shall conform to figures 1 through 4 of the specification sheet. The cover shall be fitted with a hold-down pad that functions as the cover's gasket and shall have integral projections that hold down each battery cell. The cover gasket/cell hold-down pad shall conform to figure 5 of the specification sheet. The cover pad/gasket shall be sized and fitted to match the rim of the container body for a gastight fit and effective fluid seal between the cover and the container body. Metal or wood products (such as cork) shall not be used as the gasket. If elastomers are used as the gasket, the elastomer shall conform to the requirements of 3.10.3 (see 6.10.2)."

6.2 Add the following paragraphs:

"3.7.2.1 Hold-down pad (15-1 only). The location of the projections shall be determined by the location of each intercell connector. The projections shall contact each

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intercell connector and shall apply enough force on each intercell connector to restrain vertical movement of the cells during vibration in accordance with 4.5.16 and 4.5.33. An insulating liner 0.062 ± 0.007 inch thick shall be contained between the inside, top surface of the container cover and the cover pad/gasket. The liner shall provide complete coverage of the inside, top surface of the container cover and shall conform to the requirements of 3.10.3 or 3.10.4, as applicable (see 6.10.2)."

"3.7.2.2 Hold-down pad (15-2 only). The location of the projections shall be determined by the location of each vented filler cap. The projections shall contact each vented filler cap and shall apply enough force on each vented filler cap to restrain vertical movement of the cells during vibration in accordance with 4.5.16."

6.3 Modify the following paragraphs:

3.7.5 Handles. Not applicable.

3.7.8 Receptacles. Following the last sentence, insert: "The receptacle shall be vertically positioned so that the top of the terminal block and the cell terminal contact surfaces are in the same plane. The receptacle to be installed on each battery shall be as shown on figure 11 of the specification sheet."

3.7.11.6 Vented filler cap. Following the first sentence, insert: "The vented filler cap to be installed on each cell of the battery shall be as shown on figure 11 of the specification sheet."

3.8.4.1 Limited-repair batteries. In the first sentence, change "figure 1" to "figure 9 of the specification sheet."

3.8.4.3 Battery caution and receptacle marking. In the second and third sentences, change "figure 3" to "figure 10 of the specification sheet." Insert the following after the third sentence: "The assigned CAGE code shall be inserted at the indicated area."

3.10.8 Dimensions and weights. Delete the entire text and substitute: "The dimensions and configuration of the battery container and container cover shall be as shown on figures 1 through 5 of the specification sheet. The weight of the battery shall be not greater than that shown on figure 11 of the specification sheet with the electrolyte level adjusted in accordance with 4.4.4. The base dimensions of the container shall be within specified tolerance for a distance of not less than 1.8 inches up from the bottom of the container. Beyond this distance, the case dimensions may exceed the base dimension by 50 percent of the specified tolerance."

6.4 Add the following paragraphs:

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"3.10.8.1 PIN M81757/15-1 only. All dimensions are in inches and are prior to coating except overall width (10.148 max), overall length (10.662 max), overall height (8.812 max), container base width (7.838 + 0.062, -0.15), and container base length (9.938 ± 0.062)."

"3.10.8.2 PIN M81757/15-2 only. All dimensions are in inches and are prior to coating except overall width (9.998 max), overall length (10.662 max), overall height (8.812 max), container base width (7.688 ± 0.062), and container base length (9.938 ± 0.062)."

6.5 Modify the following paragraphs:

3.10.10 Handle strength. Not applicable.

3.10.14 Capacity. Delete the first sentence and insert: "The initial capacity of the battery shall be not less than 29 Ah. The minimum capacity of the battery shall be not less than 25 Ah."

3.10.14.1 Initial capacity. To meet the initial capacity requirement, each cell shall produce a potential of not less than 1.0 volt at not less than 70 minutes into the constant 25-ampere discharge after being tested in accordance with 4.5.13 at room temperature (see 4.4.1).

3.10.15 Humidity and charge retention. In step 3.10.15.a(1), delete "60 minutes" and replace with "48 minutes."

3.10.22 Constant voltage discharge. Delete the text and substitute: "The battery shall produce not less than the current specified below after being tested in accordance with 4.5.22. The battery shall also meet the requirements of 3.9 during and after charge and discharge."

Temperature	Average current over the total 60 seconds	Current at 1 second	Current at 5 seconds	Current at 15 seconds
24.0 ± 2.0°C	650 A	925 A	825 A	700 A
-26.0 ± 2.0°C	350 A	450 A	350 A	350 A

3.10.23 Cycling. Delete the text of 3.10.23.c and insert: "c. After successfully completing the required number of duty cycles, the battery's electrical capacity shall comply with the following:

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"(1) Qualification inspection: The battery shall deliver not less than 20 Ah when subjected to 4.5.23.d. To comply with the 20-Ah requirement, the battery shall produce a potential of not less than 18.0 volts (end voltage) at not less than 60 minutes into the constant 20-ampere discharge.

"(2) Group C inspection: The battery shall deliver not less than 25 Ah when subjected to 4.5.23.d. To comply with the 25-Ah requirement, the battery shall produce a potential of not less than 20.0 volts (end voltage) at not less than 60 minutes into the constant 25-ampere discharge. The voltage of each cell shall be not less than 1.0 volt at 60 minutes into the discharge."

3.10.23 Cycling. Delete the text of 3.10.23.d and insert: "d. The battery shall successfully complete 700 duty cycles for qualification inspection and 200 duty cycles for group C inspection when subjected to 4.5.23."

3.10.25 Physical integrity at high temperature. In the first sentence, delete "and handles."

6.6 Add the following paragraphs:

"3.10.31 Slope discharge (PIN M81757/15-1 only). The battery shall perform as follows after being tested in accordance with 4.5.31.

"a. The battery shall produce a potential of not less than 12.0 volts after being discharged in accordance with figure 7 of the specification sheet for two consecutive discharges.

"b. The battery shall also meet the requirements of 3.9 during and after testing and discharge."

"3.10.32 Low frequency vibration (PIN M81757/15-1 only). The battery shall perform as follows after being tested in accordance with 4.5.32.

"a. During the charging of the battery in the test of 4.5.32, the increase in the charge current shall be not greater than 5 amperes from the lowest value reached.

"b. After being tested in accordance with 4.5.32, the battery shall produce a potential of not less than 12.0 volts when discharged in accordance with figure 8 of the specification sheet during each of the 17 vibration cycles.

"c. The battery shall also meet the requirements of 3.9 during and after testing and discharge."

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"4.4.1.1 Temperature measurement location. When the battery temperature is to be measured during laboratory testing, the temperature shall be measured at the location between the two cells shown on figure 6 of the specification sheet. The temperature shall be measured with a temperature sensor (thermocouple or thermistor) inserted between the two cells and positioned in the approximate center of the outer sidewall of each cell equidistant from top to bottom and side to side."

"4.4.10 Mounting for vibration testing (PIN M81757/15-1 only). The battery shall be clamped in the cradle by two flat bars, with one of the bars bearing on the cover adjacent to each of the left and right edges of the cover, and extending inward from the edges 0.844 ± 0.100 inch from the sides, measured in the lower 2/10 of the height, and extending from the front edge of the cover to the back edge of the cover. The hold-down bars shall be clamped to the cradle by four vertical 5/16-24 threaded rods with nuts tightened to 45 ± 3 inch-pounds, bearing on the hold-down bars through cross-beams. The rods shall be arranged symmetrically about the battery, with two rods on each side of the battery. Two of the rods shall be 2.500 ± 0.100 inches from the front of the battery, and two of the rods shall be 2.500 ± 0.100 inches from the back of the battery, measured in the lower 2/10 of the height of the battery."

6.7 Modify the following paragraphs:

4.5.9 Handle strength test. Not applicable.

4.5.23 Cycling test. Insert the following after the last sentence: "The battery shall undergo 700 duty cycles for qualification inspection and 200 duty cycles for group C inspection. The temperature near the center of the battery shall be monitored and recorded throughout this test."

4.5.23 Cycling test. Delete the text of 4.5.23.b.(1) and insert: "(1) Discharge the battery for 20 seconds through a 30-milliohm resistance for qualification inspection and a 24-milliohm resistance for group C inspection."

4.5.23 Cycling test. In the first sentence of 4.5.23.d, delete "at the C-rate" and insert a new second sentence: "The battery shall be discharged at a constant 20 amperes during qualification inspection and at a constant 25 amperes during group C inspection."

4.5.25 Physical integrity at high temperature. In step c, delete the second sentence.

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6.8 Add the following paragraphs:

"4.5.31 Slope discharge test (PIN M81757/15-1 only). This test shall consist of the following steps.

"a. Place the fully charged battery in a 38° ± 2°C (100° ± 5°F) environment and allow its temperature to stabilize.

"b. While the battery remains within the temperature environment of 4.5.31.a, discharge the battery in accordance with figure 7 of the specification sheet.

"c. Place the battery in a room temperature environment and constant potential charge the battery at 27.74 ± 0.10 volts for 5.0 ± 0.1 hours.

"d. Repeat 4.5.31.a and 4.5.31.b.

"e. Continuously measure the terminal voltage and temperature of the battery.

"f. Examine the battery for the requirements of 3.10.31."

"4.5.32 Low frequency vibration test (PIN M81757/15-1 only). The battery shall not receive any maintenance throughout this test. If the test battery's terminals are an open circuit for not less than 24 hours after starting and before completing this test, the battery shall be constant potential charged in accordance with 4.4.2 before continuing the test. This test shall consist of the following steps.

"a. Remove the cover from the battery.

"b. Constant potential charge the battery in accordance with 4.4.2.

"c. Vibration cycle: The following steps comprise one vibration cycle. Steps 4.5.32.c(3) and 4.5.32.c(4) will amount to a total of 5 hours of vibration.

"(1) Place the battery in a 38° ± 2°C (100° ± 5°F) environment and allow its temperature to stabilize. Perform the discharge of figure 7 of the specification sheet.

"(2) Remove the battery from the 38° ± 2°C (100° ± 5°F) environment, secure the cover onto the battery, and proceed immediately with 4.5.32.c(3).

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"(3) Attach the battery (oriented in its normal upright position) to the vibration equipment (see 4.4.10). Ventilate the battery through the vent tubes at 1 ± 10 percent cubic feet per minute during vibration. Vibrate the battery in accordance with figure 8 of the specification sheet. Constant potential charge the battery at 27.74 ± 0.10 volts while the battery is subjected to vibration.

"(4) Repeat 4.5.32.c(3) fourteen times.

"d. Repeat 4.5.32.c sixteen times. The seventeen vibration cycles will amount to a total of 85 hours of vibration. Step 4.5.32.c(3) shall be repeated not more than two times within a 24-hour period.

"e. Continuously measure the battery's terminal voltage and temperature and the charging current.

"f. Examine the battery for the requirements of 3.9 and 3.10.32."

6.9 Modify the following paragraphs, tables, and figures:

6.1 Intended use. Add the following after the last sentence: The M81757/15-X limited-repair battery is designed so that its cells and all components associated with installing the cells cannot be removed and replaced. The battery should be requisitioned and used only by those DoD Departments and Agencies whose aircraft battery maintenance/repair policies, procedures, and practices disallow the battery's cells to be removed and replaced."

6.10.3 Handles. Not applicable.

TABLE III. Qualification inspection of batteries. For the visual and mechanical test (test number 2), add "requirements 4 and 5" to the requirement paragraph column. For the dimensions and weight test (test number 3), add "requirement 2" to the requirement paragraph column. Delete test number 5 (Handle Strength Test). For the initial capacity discharge test (test number 8) and the humidity and charge retention test (test number 9), add "requirement 3" to the requirement paragraph column. Add the following test:

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TEST NUMBER	EXAMINATIONS AND TESTS	SAMPLE NUMBER					REQUIREMENT PARAGRAPH	TEST PARAGRAPH
	.	1	2	3	4	5		
26a	LOW FREQUENCY VIBRATION TEST (PIN M81757/15-1 only)	X					3.9, 3.10.32	4.5.32

TABLE VII. Group B inspection. Delete the table and insert the new table VII.

TABLE VII. Group B inspection.

Test No.	Examinations and Tests	Requirement Paragraph (Gen. Spec.)	Test Paragraph	Requirement Paragraph (Spec. Sheet)
	Subgroup I:			
I-1	Dimensions and weight	3.10.8, 3.10.9	4.5.8	2, 4
I-2	Cover seal test	3.8.1, 3.9,	4.5.10	
I-3	Vented filler cap test (two vented filler caps per battery)	3.10.11 3.9, 3.10.5	4.5.5.1	
	Subgroup II:			
II-1	Initial capacity discharge	3.10.14	4.5.13	3
II-2	20-second pulse discharge	3.9, 3.10.20	4.5.20	
	Subgroup III:			
III-1	Constant voltage discharge	3.9, 3.10.22	4.5.22	--
III-2	Internal pressure test	3.9, 3.10.26	4.5.26	

TABLE VIII. Group C inspection. For the visual and mechanical test (test numbers I-2, II-1, and III-1), add "requirements 4 and 5" to the requirement paragraph column. For the dimensions and weight test (test number I-3), add "requirement 2" to the requirement paragraph column. For the initial capacity discharge tests (test numbers I-6, II-5, and III-4)

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and the humidity and charge retention test (test number II-6), add "requirement 3" to the requirement paragraph column. Add the following test:

TEST NO.	EXAMINATIONS AND TESTS	REQUIREMENT PARAGRAPH	TEST PARAGRAPH
	SUBGROUP I:		
I-6a	SLOPE DISCHARGE TEST (PIN M81757/15-1 only)	3.9, 3.10.31	4.5.31

FIGURE 1. Marking for limited-repair batteries. Delete the figure and insert figure 9 herein.

FIGURE 3. Marking for battery caution and receptacle. Delete the figure and insert figure 10 herein.

CONCLUDING MATERIAL

Custodian:
Navy - AS

Preparing activity:
Navy - AS

Agent:
Navy - NW
(Project 6140-N888)

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